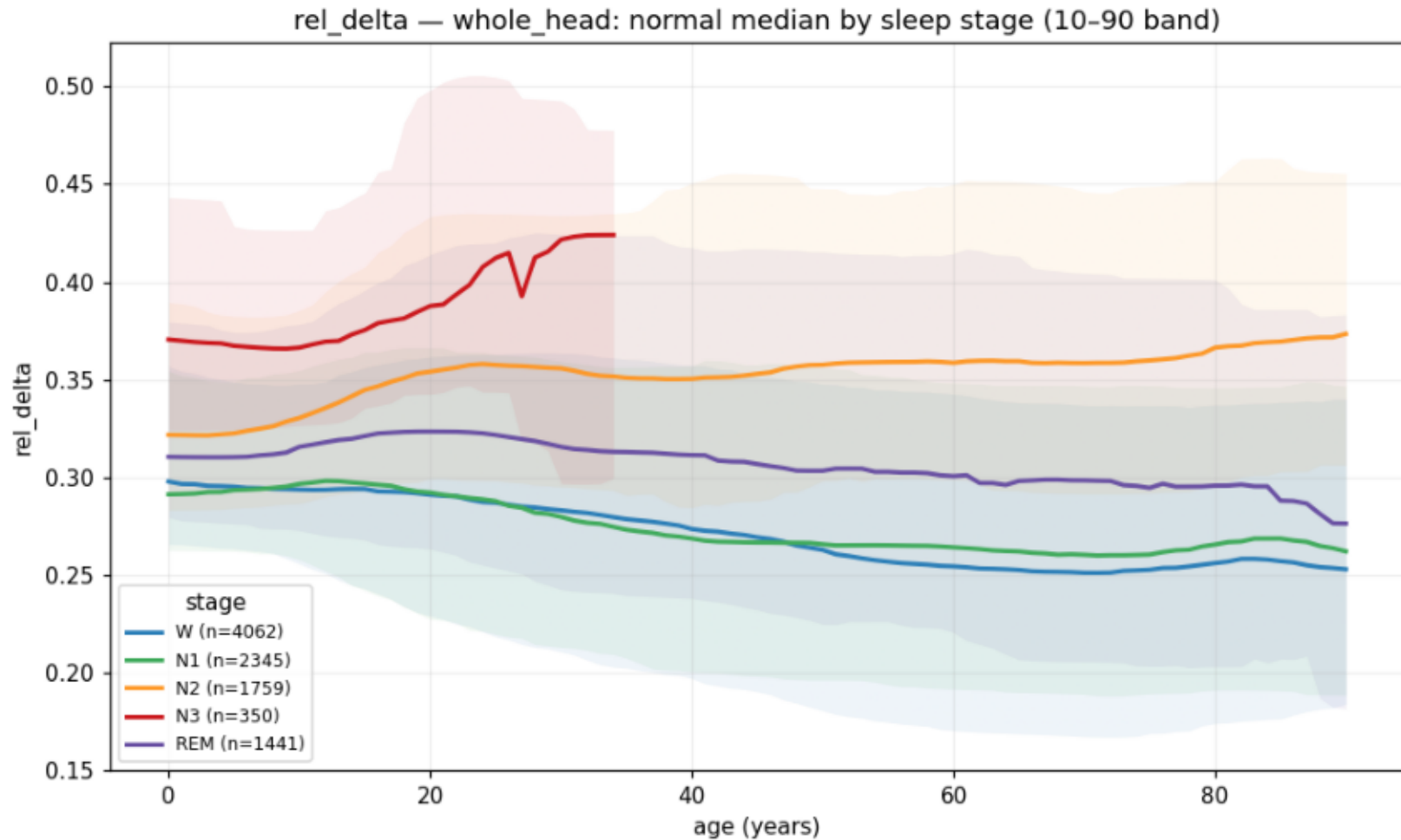


EEG slowing: growth curves & sleep-stage norms

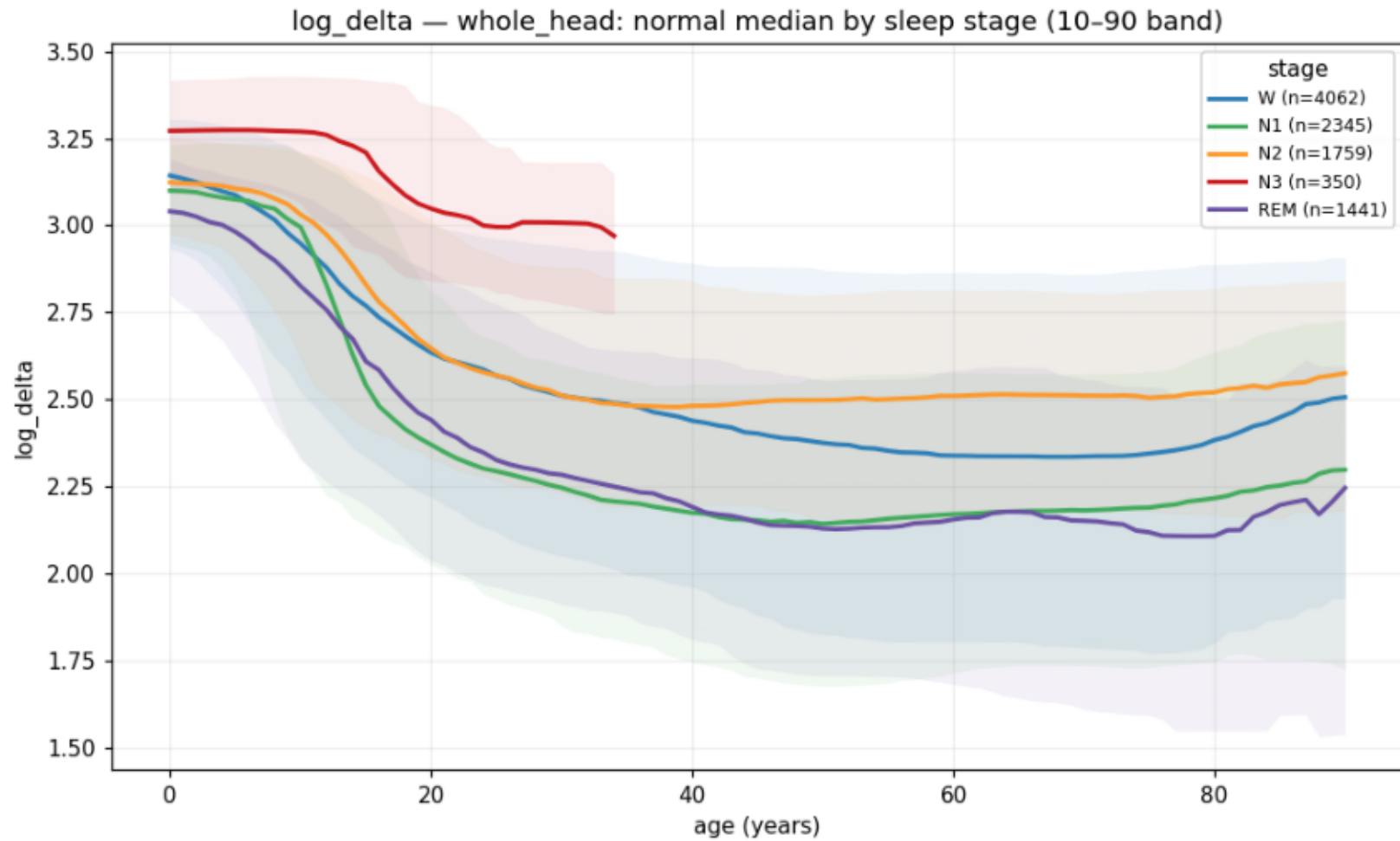
12,379 recordings, sleep-staged with morgoth2. Normal delta rises with sleep depth ($N3 > N2 > W$); each segment scored vs its own stage's normal curve, so physiological deep-sleep delta is not called abnormal. Descriptions add prevalence, persistence, and stage-accentuation.

Relative delta by sleep stage (v2)



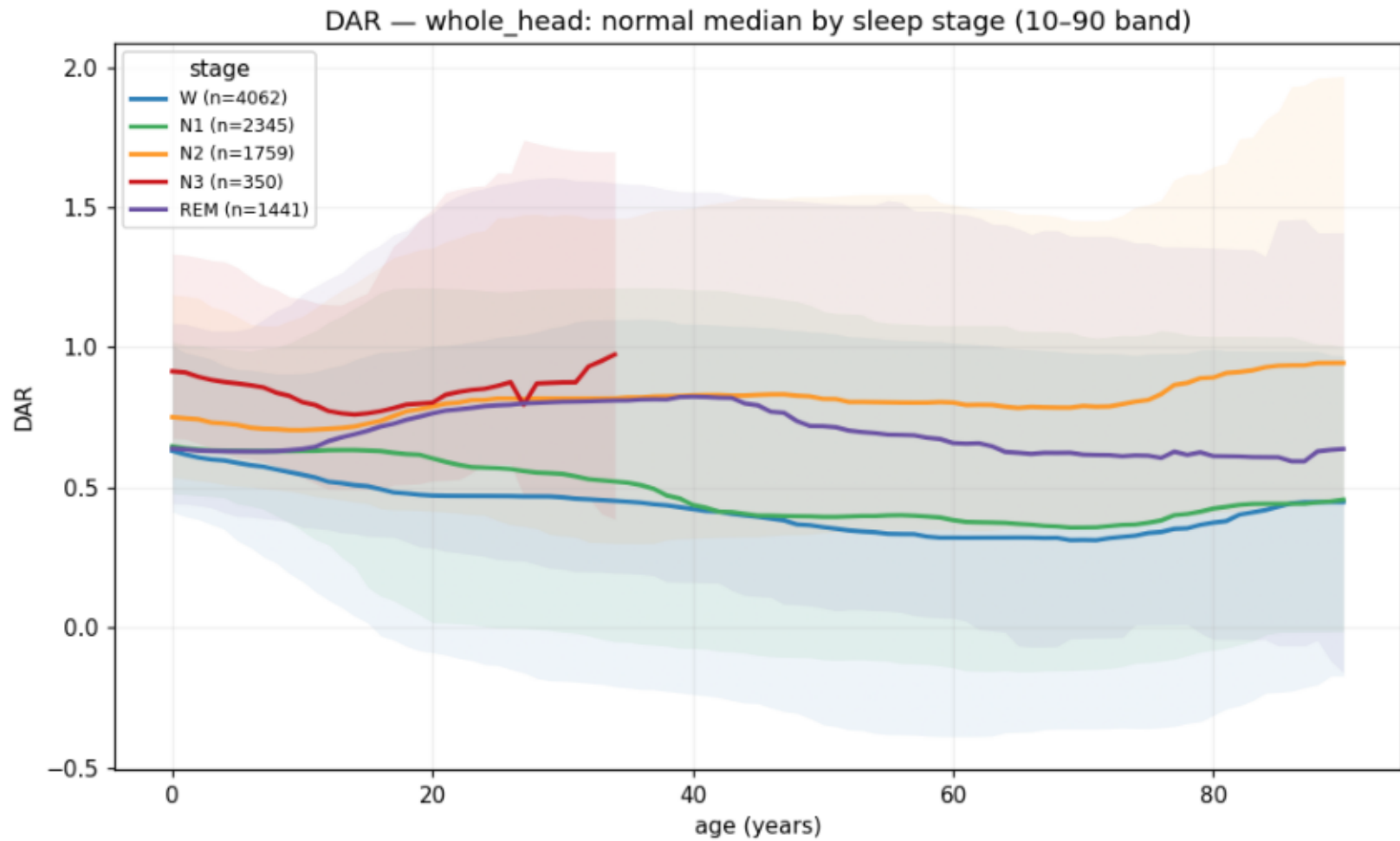
Normal relative delta rises with sleep depth: W~N1 < N2 < N3 (REM intermediate); N3 truncates ~age 34. The same delta abnormal in wake is normal in N2/N3 -> scoring uses the stage-matched curve.

Absolute delta by sleep stage (v2)



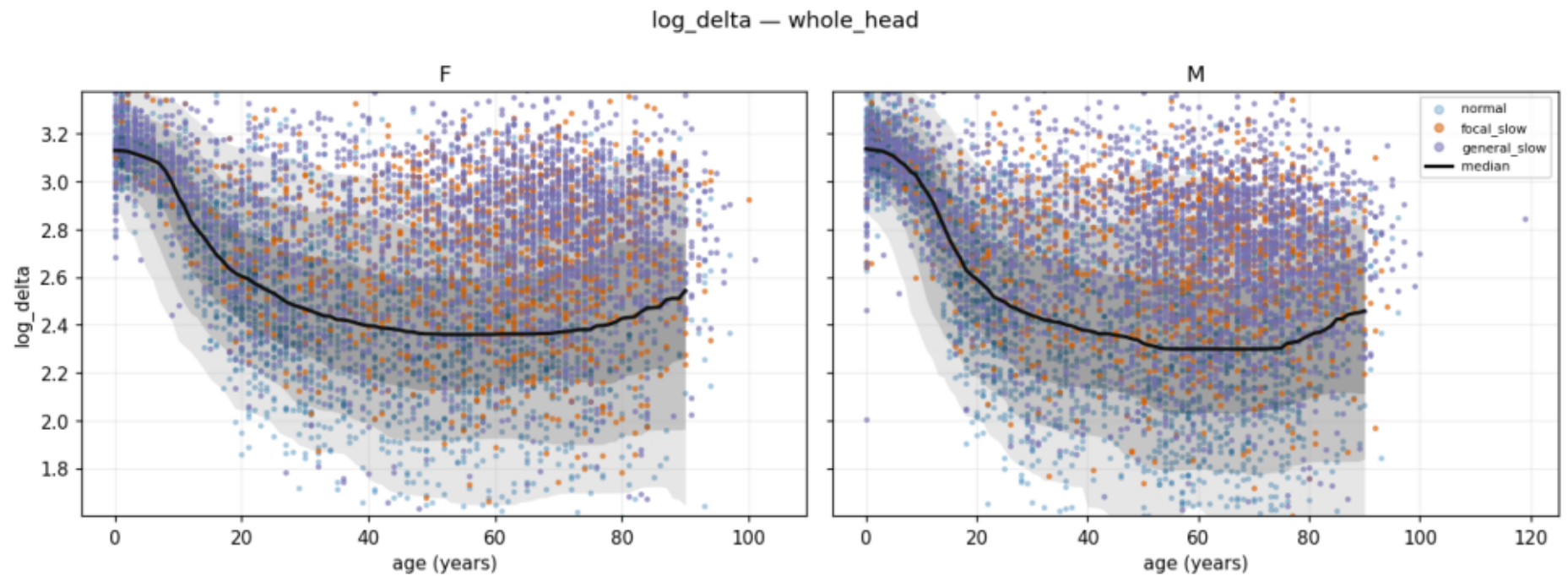
Deep sleep (N3) highest, wake lowest, across age.

Delta/alpha ratio by sleep stage (v2)



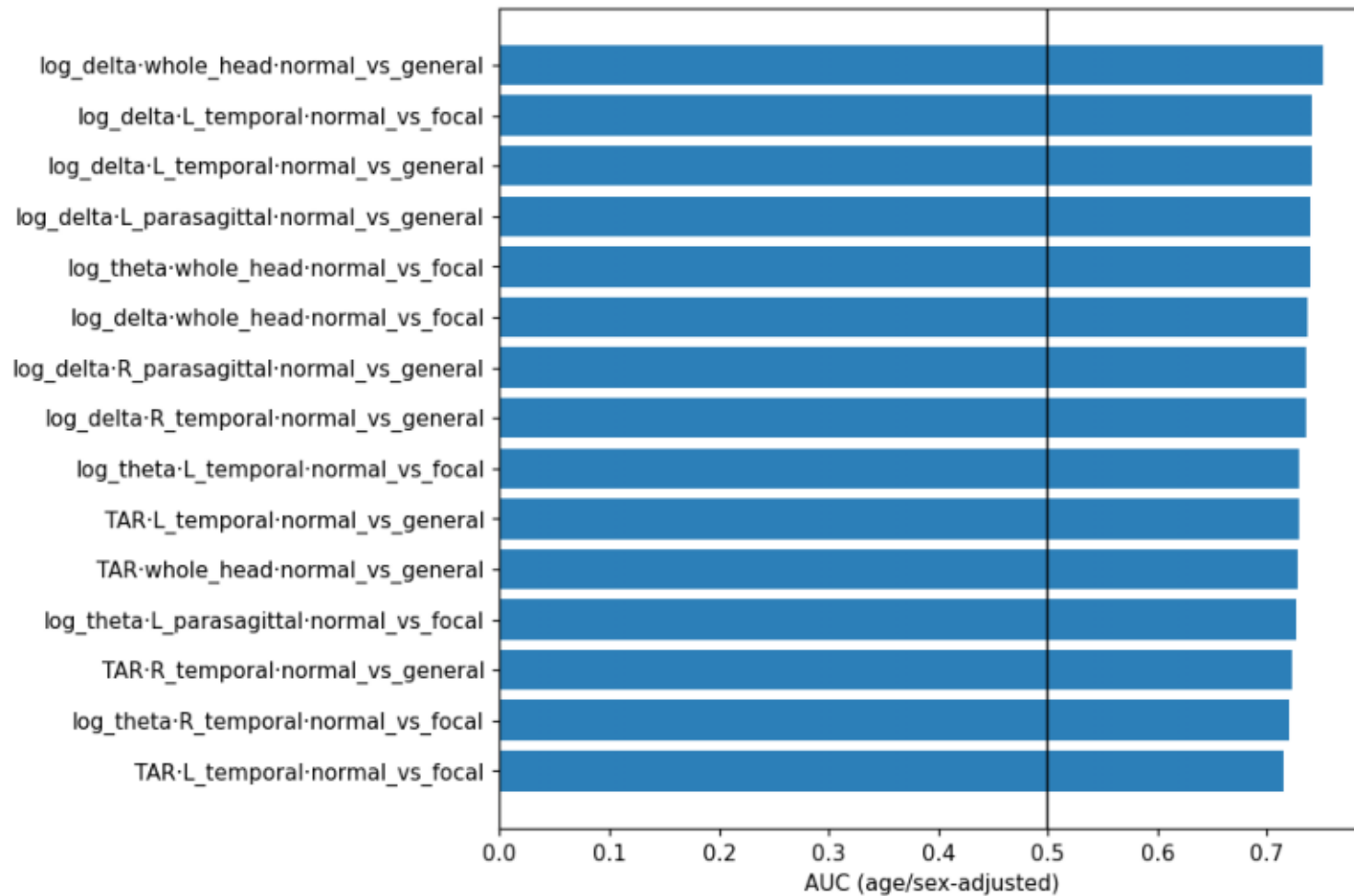
DAR rises in deeper sleep; the reference must be stage-specific.

Absolute delta - development (stage-pooled)



Normal delta falls through childhood to a plateau by ~30; slowing groups sit above the band.

Discriminative features (age/sex-adjusted AUC)



Absolute delta/theta power and TAR lead (AUC 0.73-0.75).

Example report sentences (stage-aware)

Focal-slowing recordings

- (age 49 F) Abundant moderate generalized delta slowing, present in 89% of segments, accentuated in N1; peak 3.9 SD above stage-matched norms; longest run 3.8 min over 3 episodes.
- (age 82 F) Abundant mild generalized delta slowing, present in 63% of segments, accentuated in N1; peak 2.9 SD above stage-matched norms; longest run 3.8 min over 2 episodes.
- (age 72 M) Frequent moderate generalized delta slowing, present in 32% of segments, present only during sleep, accentuated in N1; peak 3.1 SD above stage-matched norms; longest run 2.4 min over 2 episodes.
- (age 81 F) Abundant mild generalized delta slowing, present in 51% of segments, present only during sleep, accentuated in N2; peak 2.7 SD above stage-matched norms; longest run 1.9 min over 7 episodes.

Generalized-slowing recordings

- (age 23 F) Frequent moderate generalized delta slowing, present in 45% of segments; peak 3.5 SD above stage-matched norms; longest run 3.3 min over 17 episodes.
- (age 2 M) Abundant moderate generalized delta slowing, present in 55% of segments, accentuated in N3; peak 3.0 SD above stage-matched norms; longest run 1.2 min over 22 episodes.
- (age 24 F) Frequent moderate generalized delta slowing, present in 44% of segments, accentuated in REM; peak 3.6 SD above stage-matched norms; longest run 1.2 min over 24 episodes.
- (age 43 F) Abundant mild generalized delta slowing, present in 60% of segments, accentuated in REM; peak 2.9 SD above stage-matched norms; longest run 0.9 min over 14 episodes.